B. Tournament

time limit per test: 2 seconds
memory limit per test: 256 megabytes

The tournament «Sleepyhead-2010» in the rapid falling asleep has just finished in Berland. n best participants from the country have participated in it. The tournament consists of games, each of them is a match between two participants. $n \cdot (n - 1) / 2$ games were played during the tournament, and each participant had a match with each other participant.

The rules of the game are quite simple — the participant who falls asleep first wins. The secretary made a record of each game in the form «x_i y_i», where x_i and y_i are the numbers of participants. The first number in each pair is a winner (i.e. x_i is a winner and y_i is a loser). There is no draws.

Recently researches form the «Institute Of Sleep» have found that every person is characterized by a value p_j — the speed of falling asleep. The person who has lower speed wins. Every person has its own value p_j, constant during the life.

It is known that all participants of the tournament have distinct speeds of falling asleep. Also it was found that the secretary made records about all the games except one. You are to find the result of the missing game.

Input

The first line contains one integer n $(3 \le n \le 50)$ — the number of participants. The following $n \cdot (n - 1) / 2 - 1$ lines contain the results of the games. Each game is described in a single line by two integers x_i, y_i $(1 \le x_i, y_i \le n, x_i \ne y_i)$, where x_i и y_i are the numbers of the opponents in this game. It is known that during the tournament each of the n participants played $n - 1$ games, one game with each other participant.

Output

Output two integers x and y — the missing record. If there are several solutions, output any of them.

Examples

input Copy

```
4
4 2
4 1
2 3
2 1
3 1
```

output Copy

```
4 3
```

→ **Attention**

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 27 (Codeforces format, Div. 2)

Finished

Practice

→ **Virtual participation**

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ **Clone Contest to Mashup**

You can clone this contest to a mashup.

Clone Contest

→ **Submit?**

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ **Last submissions**

Submission	Time	Verdict
372417827	Apr/25/2026 09:01	Accepted
372417271	Apr/25/2026 08:54	Accepted
372417081	Apr/25/2026 08:52	Accepted

→ Problem tags

bitmasks | brute force | dfs and similar | greedy | *1300

No tag edit access

→ Contest materials

- Announcement
- Tutorial #1
- Tutorial #2 (en)
- Tutorial #3 (en)

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